

POE Quick Review (5th BPE)

Module – A

1. Microeconomics and Macroeconomics Are Interdependent and Complement Each Other

- **Microeconomics:** Focuses on individual decision-making and behavior.
- **Macroeconomics:** Examines the economy as a whole, like national income and unemployment.
- **Interdependence:** Changes in macroeconomics impact microeconomics and vice versa.
- **Impact:** Economic factors like inflation can influence business decisions, and individual choices affect national economic performance.

2. Positive and Normative Economics

- **Positive Economics:** Focuses on facts and objective analysis.
- **Normative Economics:** Involves opinions about what should be done in the economy.
- **Positive:** Deals with statements based on real data.
- **Normative:** Offers suggestions based on ethical or societal values.

3. "Scarcity Brings in the Problem of Choice" – Explanation

- **Scarcity:** Resources are limited, forcing choices.
- **Decision-making:** People must prioritize their needs based on limited resources.
- **Trade-offs:** Every choice involves giving up something else.
- **Impact on Prices:** Scarcity increases demand and can raise prices.

4. Opportunity Cost

- **Definition:** The value of the next best alternative when making a decision.
- **Decision-making:** Helps evaluate options by considering what's given up.
- **Resource Management:** Guides efficient use of resources by comparing alternatives.
- **Economic Growth:** Businesses use opportunity cost to choose investments that provide the best return.

5. Four Major Economic Factors for Production of Goods and Services

- **Land:** Natural resources required for production.
- **Labor:** Human effort used in producing goods and services.
- **Capital:** Machinery and tools used in production.
- **Entrepreneurship:** Individuals who combine resources to create businesses.

6. Production Possibility Curve (PPC)

- **PPC Definition:** A graph showing potential production combinations of two goods.
- **Efficiency:** Points on the curve represent the best use of resources.
- **Technological Advancement:** Technology can shift the curve outward, increasing production.
- **Graphical Impact:** PPC shows the trade-off between two goods in the economy.

7. Economic Model

- **Definition:** An economic model helps understand real economic events using theories and equations.
- **Supply and Demand Model:** Explains the relationship between price and quantity of a product.

- **Monetary Policy Model:** Describes how central banks use interest rates to control economic activity.
- **Impact:** Economic models help policymakers make informed decisions.

8. Circular Flow Model

- **Households:** Supply labor and consume goods and services.
- **Businesses:** Produce goods, hire labor, and sell products to households.
- **Government:** Collects taxes and provides public services.
- **Foreign Sector:** Represents exports and imports with foreign countries.
- **Money Flow:** Households pay for goods and services; businesses pay wages.

9. Market Economy vs. Controlled Economy

Characteristic	Market Economy	Controlled Economy
Ownership of Resources	Resources are owned by private individuals and businesses.	Resources are owned and controlled by the government.
Decision-Making	Decisions are driven by supply and demand.	Government determines production, pricing, and allocation.
Role of Government	Minimal interference from the government.	Government plays a central role in the economy.
Profit and Incentives	Profit-driven, with businesses competing for market share.	Focus on social welfare, with limited incentives for profit.
Efficiency & Consumer Choice	High efficiency and variety in goods and services.	Lower efficiency with fewer choices for consumers.

10. Difference between Market and Command Economy

Characteristic	Market Economy	Command Economy
Ownership of Resources	Private individuals and businesses own resources.	Government owns and controls resources.
Decision-Making	Determined by supply and demand.	Government decides production and distribution.
Role of Government	Minimal government interference.	High government control over the economy.
Profit and Incentives	Businesses operate for profit; competition exists.	Focus on social welfare; limited profit motive.
Efficiency & Consumer Choice	High efficiency and variety of choices.	Less efficiency, limited consumer choices.

11. Point inside the Production Possibilities Frontier (PPF)

- **Feasible but Inefficient:** A point inside the PPF is possible but represents underutilization of resources.
- **Efficiency:** Efficiency means using all resources to their full potential.
- **Example:** A factory producing less than its potential output shows inefficiency.
- **Conclusion:** Points inside the PPF are feasible but not efficient.

12. Point on the Production Possibilities Curve (PPC)

- **Feasible and Efficient:** A point on the PPC shows maximum output with full resource use.
- **No Waste:** Resources are fully utilized at points on the curve.
- **Example:** Producing 100 units of food and 50 units of clothing maximizes output.
- **Conclusion:** A point on the PPC is both feasible and efficient.

13. Circular-Flow Diagram and Factor Markets

- **Agreement:** Households are sellers, and firms are buyers in factor markets.
- **Households as Sellers:** They provide labor and other factors of production to firms.
- **Firms as Buyers:** Firms purchase factors of production (labor, land, capital) from households.
- **Example:** A household sells labor to a firm for wages, and firms use labor to produce goods.
- **Conclusion:** The circular-flow diagram accurately represents the exchange of resources in the economy.

Module – B

1. Law of Supply

- **Definition:** The Law of Supply states that as the price of a product increases, its supply increases, and as the price decreases, the supply decreases (ceteris paribus).
- **Figure Description:**

Price (Taka) | Supply (Units) -> (1 | 5), (2 | 10), (3 | 15), (4 | 20), (5 | 25)

- **Key Reason:** Producers maximize profit by supplying more when prices are high and less when prices are low.

2. Change in Supply and Change in Quantity Supplied

- **Change in Supply:** It is the increase or decrease in supply at the same price due to external factors (production costs, technology, etc.). The supply curve shifts.
 - **Figure Description:**
 - Price OP, Seller moves from point A on SS curve (OQ1) to point B on S1S1 curve (OQ2) due to external factors.
- **Change in Quantity Supplied:** It is a change in the quantity supplied due to price changes, with other factors constant. Movement occurs along the same supply curve.
 - **Figure Description:**
 - Price increases from P1 to P2, and quantity supplied expands from Q1 to Q2. Price decreases from P2 to P1, quantity supplied contracts from Q2 to Q1.

3. Absence of Supply Curve in a Monopoly

- **Reason:** In a monopoly, the single seller controls the market price, eliminating a conventional supply curve. The monopolist adjusts supply based on demand and market control.

4. Demand and Supply Diagram

- **Definition:** The equilibrium price is where the demand and supply curves intersect. The equilibrium quantity is where both curves meet, and total revenue is the product of equilibrium price and quantity.
- **Figure Description:**
 - At the intersection of demand (D) and supply (S) curves, equilibrium price (P) and quantity (Q) are determined, influencing total revenue ($P \times Q$).

5. Change in Demand vs. Change in Quantity Demanded

- **Change in Demand:** Caused by factors like income, preferences, or market expectations, shifting the demand curve.
 - **Figure Description:**

- Price remains constant, but quantity demanded increases from Q to Q2 due to external factors, shifting the demand curve from DD to D'D'.
- **Change in Quantity Demanded:** It occurs due to a price change, with the demand curve remaining the same. Movement along the curve occurs.
 - **Figure Description:**
 - As price moves from P to P1, quantity demanded moves from Q to Q1, reflecting a change in quantity demanded along the DD curve.

6. Impact on Equilibrium When Supply Changes

- **Increase in Supply:** Shifts the supply curve to the right, lowering price and increasing quantity.
 - **Figure Description:**
 - The supply curve shifts from S to S1, resulting in lower price and higher quantity at the new equilibrium.
- **Decrease in Supply:** Shifts the supply curve to the left, raising price and decreasing quantity.
 - **Figure Description:**
 - The supply curve shifts from S to S2, leading to higher price and lower quantity at the new equilibrium.

Absolutely! Here's your **updated version** of questions 7 to 12 with **clear definitions** and **basic related explanations** in **easy language**, so it's easier to understand and remember:

7. Supply Schedule vs Supply Curve

Definition:

- **Supply Schedule:** A table that shows how much of a good a producer is willing to supply at different prices.
- **Supply Curve:** A graph that shows the relationship between price and quantity supplied.

Explanation:

- The supply curve is drawn from the supply schedule.
- It **slopes upward** because as price increases, producers are more willing to supply more.

Reasons for Upward Slope:

1. **Profit Incentive** – Higher price means more profit, so supply increases.
2. **Increased Costs** – At higher prices, producers can cover extra costs.
3. **Resource Allocation** – Firms shift resources to goods that bring more profit.

8. Equilibrium in the Loan Market

Definition:

- **Loan Market Equilibrium:** The point where the amount of loans people want to borrow equals the amount banks are willing to lend at a certain interest rate.

Explanation:

- At equilibrium, **loan demand = loan supply**.
- Interest rate adjusts to reach this balance.

Shortage Signs in Loan Market:

1. More borrowers competing for limited loans.
2. Interest rates start rising.
3. Banks become stricter in giving loans.
4. Loan approvals are delayed.
5. People look for alternative financing (e.g., informal lenders).

9. Indifference Curve and Indifference Map

Definitions:

- **Indifference Curve:** A curve showing combinations of two goods that give the same satisfaction to a consumer.
- **Indifference Map:** A set of indifference curves representing different levels of satisfaction.

Basic Concepts:

- Consumers are **indifferent** between any two points on the same curve.
- Higher curve = higher utility.

Characteristics of Indifference Curve:

1. **Downward Sloping** – More of one good means less of the other.
2. **Infinite Curves** – Many combinations give different satisfaction.
3. **Non-intersecting** – Curves can't cross each other.
4. **Negative Slope (MRS)** – Shows the rate of trade-off between goods.

10. Budget Line

Definition:

- **Budget Line:** A line showing all possible combinations of two goods a consumer can buy with a given income and prices.

Explanation:

- It shows the consumer's spending limit.
- If income or price changes, the budget line shifts.

Graph Description:

- **X-axis:** Quantity of Good X
- **Y-axis:** Quantity of Good Y
- **Points A, B, C:** Different combinations consumer can afford
- **Iso-cost Line:** Another name for the budget line showing same total cost

11. Convexity of Indifference Curve

- **Convex to the Origin:** The curve bends toward the origin, showing that the Marginal Rate of Substitution (MRS) decreases.

Explanation:

- As a consumer gets more of good Y, they are willing to give up **less of good X**.
- Shows the **diminishing marginal utility** of each good.

12. Indifference Curve Slopes Downwards

- **Sloping Downward to the Right:** Means if the quantity of one good increases, the other must decrease to keep utility constant.

Explanation:

- If you get more of one product, you must give up some of the other to stay equally happy.
- This reflects the **trade-off between goods**.

Graph Idea:

- Move from one point to another on the curve → more of one good, less of the other → same level of satisfaction.

13. Relationship between Indifference Curve and Marginal Rate of Substitution (MRS)

Definition:

- **Indifference Curve:** A graph showing different combinations of two goods that give the same level of satisfaction to a consumer.
- **Marginal Rate of Substitution (MRS):** The rate at which a consumer is willing to give up one good to get more of another good, keeping satisfaction the same.

Explanation:

- The curve **slopes downward** because if you get more of one good, you must give up some of the other to stay equally happy.
- The curve is **convex to the origin**, showing that MRS **decreases**—you give up less of one good as you get more of it.

14. How Total Satisfaction is Measured Using Total Utility (TU)

Definition:

- **Total Utility (TU):** The total amount of satisfaction a person gets from consuming all units of a good.
- **Marginal Utility (MU):** The additional satisfaction from consuming one extra unit.

Explanation:

- As more units are consumed, **TU increases**, but the rate of increase **slows down** (MU decreases).
- When MU becomes zero, **TU is at maximum**.
- $TU = \text{Sum of all MU}$.

15. Relation between Total Utility (TU) and Marginal Utility (MU)

Definition:

- **Total Utility (TU):** Total satisfaction from all units.
- **Marginal Utility (MU):** Extra satisfaction from one more unit of the good.

Explanation:

- TU increases as long as MU is positive.
- When $MU = 0 \rightarrow$ TU is at its highest.
- If MU is negative $\rightarrow$ TU starts to fall.
- **$MU = \text{Change in TU} / \text{Change in Quantity}$**

16. Perfectly Inelastic Supply Curve

Definition:

- **Perfectly Inelastic Supply:** A situation where the quantity supplied does **not change** no matter how much the price changes.

Explanation:

- This happens in the **short run** for rare goods (like original paintings).
- The **supply curve is vertical**, meaning supply is **fixed**.

17. Income Elasticity of Demand & Luxury Goods

Definition:

- **Income Elasticity of Demand:** It shows how much demand changes when a person's **income changes**.
- **Luxury Goods:** Goods for which demand increases **more than proportionally** as income increases.

Explanation:

- Income Elasticity > 1 for luxury goods (like air conditioners, vacations).
- These goods are not essential but are demanded more as people get richer.

18. Price Increase & Total Revenue (Elastic Demand)

Definition:

- **Elastic Demand:** When a small change in price causes a big change in quantity demanded.
- **Total Revenue (TR):** Total money a seller gets ($\text{Price} \times \text{Quantity Sold}$).

Explanation:

- In case of elastic demand:
 - If price $\uparrow \rightarrow$ quantity $\downarrow$ a lot $\rightarrow$ **TR falls**
 - If price $\downarrow \rightarrow$ quantity $\uparrow$ a lot $\rightarrow$ **TR rises**

Example:

- Price \$10, Quantity 100 $\rightarrow$ TR = \$1000
- Price \$12, Quantity 70 $\rightarrow$ TR = \$840 $\rightarrow$ TR decreased

19. Why Consumer Consumes Until Marginal Utility = Price

Definition:

- **Marginal Utility (MU):** Extra satisfaction from consuming one more unit.
- **Price (P):** Cost of one unit of the product.

Explanation:

- A consumer gets maximum satisfaction when $MU = \text{Price}$.
 - If $MU > P \rightarrow$ extra satisfaction $\rightarrow$ buy more
 - If $MU < P \rightarrow$ not worth the price $\rightarrow$ buy less
 - So, the consumer stops at $MU = P$ to balance cost and benefit.

Module – C

1. Limitations of the Law of Diminishing Marginal Utility

Definition:

This law states that as a person consumes more units of a good, the additional satisfaction (utility) from each extra unit decreases.

Limitations:

1. **Irreversible Conditions:** When a person is in extreme need (e.g., very thirsty), the first few units may give increasing utility.
2. **Variety and Change:** If goods are varied (like eating different dishes), utility might not diminish.
3. **Mental State:** A person's emotional or psychological state (like strong preference) can maintain or increase utility.
4. **Scarcity of Resources:** When something is rare or critical (like medicine), each unit may have high utility.
5. **Combination of Goods:** When two goods are consumed together (like food and drink), the law may not apply as expected.

2. Relationship between Total Cost, Average Cost, and Marginal Cost

Definitions:

- **Total Cost (TC):** All costs incurred in production (Fixed Cost + Variable Cost).
Formula: $TC = FC + VC$
- **Average Cost (AC):** Cost per unit of output.
Formula: $AC = TC / Q$
- **Marginal Cost (MC):** Cost of producing one extra unit.
Formula: $MC = \Delta TC / \Delta Q$

Relationship:

1. **AC depends on TC** – as TC increases, AC may rise or fall based on output level.
2. **MC and AC Relationship:**
 - If $MC < AC$, then **AC falls**.
 - If $MC > AC$, then **AC rises**.
 - If $MC = AC$, **AC is at its minimum**.

Graph Description:

- **AC and MC** are both U-shaped.
- **MC** cuts **AC** at its minimum point.

3. 'MC and AC curves intersect where AC is minimum' – Explained

Correct Statement:

The **MC** curve intersects the **AC** curve at the **lowest point of AC**, not where MC is maximum.

Explanation:

- **If $MC < AC$:** AC is decreasing.
- **If $MC > AC$:** AC is increasing.
- **If $MC = AC$:** AC is at its minimum.

Why It Happens:

- Think of MC like an “influencer” of AC.
- If the extra cost is lower than average, the average goes down.
- If the extra cost is higher, the average goes up.

Graph Description:

- AC and MC are U-shaped curves.
- MC curve intersects AC curve exactly at its lowest point.

4. Isoquant ap

Definition:

An isoquant map shows different **isoquants**, each representing a **different level of output** using different combinations of inputs (e.g., labor and capital).

Key Features:

- **Each isoquant = fixed output level.**
- **More isoquants above = higher output.**
- Shows **input substitution** while maintaining the same output.
- Helps firms find the **most efficient input mix**.

Graph Description:

- X-axis: Labor, Y-axis: Capital.
- Curves: IQ1, IQ2, IQ3 (higher isoquant = more output).

5. Why Isoquants Are Convex

Main Reason:

- **Diminishing Marginal Rate of Technical Substitution (MRTS)** – means as one input (labor) increases, less of the other (capital) is needed, but at a decreasing rate.

Reasons:

1. **Diminishing MRTS:** More labor replaces less capital.
2. **Law of Diminishing Returns:** Extra input gives less extra output over time.
3. **Input Nature:** Inputs like labor and capital aren't perfect substitutes.

Graph Description:

- Convex curves bending **towards the origin**.
- Slope flattens as you move right – shows diminishing MRTS.

Special Shapes:

- **Straight Line:** Perfect substitutes.
- **L-Shape:** Perfect complements (used in fixed proportion).

6. Profit Maximization Condition: $MC = MR$ and MC cuts MR from below

Definitions:

- **MC (Marginal Cost):** Cost of producing one more unit.
- **MR (Marginal Revenue):** Revenue from selling one more unit.

Profit Maximization Rule:

- A firm maximizes profit when **$MC = MR$** .
- But it's only true if **MC cuts MR from below**, i.e., when MC is rising.

Why This Condition?

- If **$MC < MR$** , produce more (each unit gives profit).
- If **$MC > MR$** , reduce output (each unit gives loss).
- Profit is highest when **$MC = MR$** and is **rising**.

Graph Description:

- X-axis: Quantity, Y-axis: Cost & Revenue.
- MR is a horizontal line (for perfect competition).
- MC is U-shaped and cuts MR from below at the profit-maximizing point.

Module – D

1. Monopolistic Competition

- **Definition:** Many firms sell similar but slightly different products.
- **Many Sellers:** Multiple firms compete in the same market.
- **Product Differentiation:** Each firm's product is slightly unique (e.g., clothing brands).
- **Limited Price Control:** Firms can set prices within a small range.
- **Free Entry/Exit:** Firms can enter or leave the market freely.
- **Graph:** Downward-sloping demand curve; profit in short run, normal profit in long run.
- **Example:** Local restaurants offering different tastes and ambiance.

2. Collusion and Cartels in Oligopoly

- **Definition:** Firms agree to fix prices or limit output to control the market.
- **Incentive to Cheat:** Firms may secretly break the agreement for more profit.
- **Legal Barriers:** Cartels are illegal under competition laws (e.g., Bangladesh Competition Act).
- **New Entrants:** New firms disrupt the cartel's control.
- **Demand Changes:** Falling demand causes internal disputes.
- **Personal Gain:** Firms prioritize their own profit, weakening the cartel.
- **Example:** OPEC often fails to maintain oil production limits.

3. Oligopoly Market

- **Definition:** A few large firms dominate the market.
- **Limited Competition:** Each firm's decisions influence others.
- **Homogeneous/Differentiated Products:** Products may be similar (steel) or branded (cars).
- **Price Influence:** Firms react to each other's pricing strategies.
- **Cartel Formation:** Firms may collude to limit competition.
- **Graph (optional):** Kinked demand curve if pricing behavior is strategic.

4. Absence of Supply Curve in Monopoly

- **Single Seller:** One firm controls the whole market.
- **Price Setter:** Firm sets prices based on demand, not supply.
- **No Entry:** Barriers prevent competition.
- **Centralized Control:** Output depends on profit-maximizing decision, not a supply curve.
- **Graph:** Only a demand curve and marginal cost/marginal revenue curve; no supply curve.

5. Accounting Profit vs. Economic Profit

- **Accounting Profit:** Revenue minus explicit costs (e.g., wages, rent).
- **Economic Profit:** Revenue minus both explicit and implicit costs (e.g., opportunity cost).
- **Example:**
 - Accounting Profit = \$100,000 – \$70,000 = \$30,000
 - Economic Profit = \$100,000 – (\$70,000 + \$20,000) = \$10,000

6. Perfectly Competitive Market & Break-Even Point

Perfectly Competitive Market:

1. **Many buyers and sellers** – No one can set the price; they follow the market price.
 2. **Easy to enter or leave** – Any business can start or stop selling anytime.
 3. **Same kind of products** – All sellers sell the same product.
 4. **Full information** – Buyers and sellers know everything about price and quality.
- Example:** Rice or salt market in Bangladesh.

Break-Even Point:

1. **Fixed cost** – Cost that stays the same (like rent).
2. **Variable cost** – Cost that changes with production (like raw materials).
3. **Total cost** – It's fixed cost + variable cost.
4. **Break-even point** – When total cost = total revenue (no profit, no loss).

7. Allocative Efficiency in Perfect Competition

1. **Price = cost of last unit ($P = MC$)** – Firms produce the right amount for society.
2. **No waste** – Firms don't produce too much or too little.
3. **Maximum benefit** – Both buyers and sellers are satisfied.
4. **Free entry/exit** – Keeps prices and output balanced.
5. **Resources used properly** – No overuse or underuse of anything.

8. Why Average Revenue is Lowest in Perfect Competition

1. **Firms follow market price** – They can't charge more than the market price.
2. **Price = average revenue (AR)** – So, AR stays low.
3. **No power to set price** – They must sell at the price everyone else is using.
4. **Other markets can charge more** – Monopolies or oligopolies have power to raise prices.
5. **Good for buyers** – Lower AR means lower prices and more efficiency.

9. Monopolistic vs Oligopoly Market

Monopolistic Market:

1. **Only one seller** – One company controls the market.
2. **Unique product** – No other similar product.
3. **Can set high prices** – Seller decides the price.
4. **Hard to enter** – Other firms can't easily join.
5. **Has full power** – The seller controls the whole market.

Oligopoly Market:

1. **Few big sellers** – Only a few large companies exist.
2. **Products may differ** – Products can be the same or slightly different.
3. **Can set prices but carefully** – They watch what others are doing.
4. **Hard to enter** – Not easy for new companies to join.
5. **Firms depend on each other** – One firm's action affects others.

Main difference: Monopoly has one seller, oligopoly has few big sellers.

10. Deadweight Loss in Monopol

1. **Monopoly charges higher price** – Sells less at a higher price.
2. **Customers lose** – Some people can't buy because price is too high.
3. **Monopoly earns more** – But total benefit to society is lower.
4. **Deadweight loss** – The benefit that society loses.
5. **Less efficiency** – Some buyers and sellers miss out on good deals.

Module – E

1. Market Failure

- **Market failure** happens when the market can't efficiently allocate resources, causing loss of welfare.
- **Main causes of market failure:**
 1. **Externalities:** Activities that affect others without adjusting prices (positive or negative).
 2. **Lack of information:** Consumers/producers can't make good decisions due to missing information.

3. **Infrastructure issues:** Market failure in monopolies or oligopolies.
4. **Economic power:** Some firms control prices, reducing market efficiency.

2. Positive Externality & Corrective Subsidy

- **Positive externality:** Benefits created for others without compensation (e.g., education, vaccination).
- **Corrective Subsidy:** Government provides financial support to encourage beneficial actions that create positive externalities.
 - **How it works:**
 1. **Encourage more actions** (e.g., subsidies for education or vaccination).
 2. **Increase social benefits** (e.g., healthier society).
 3. **Correct production:** Government supports companies to create more social benefits (e.g., eco-friendly tech).
- **Examples:** Education subsidies → skilled workforce → economic growth.

3. Positive and Negative Externalities

- **Positive externality:** Benefits for others (e.g., education, vaccination).
- **Negative externality:** Harm to others (e.g., pollution, smoking).
 - **Positive example:** Education improves work efficiency, reduces crime.
 - **Negative example:** Pollution harms health, increases diseases.

4. Efficient Tax System

- **Benefits of an efficient tax system:**
 1. **Increases government revenue:** Funds public services like education, healthcare.
 2. **Reduces inequality:** Progressive taxes help lower the wealth gap.
 3. **Encourages investment:** Fair taxes attract businesses and investment.
 4. **Controls inflation:** Prevents excess money in circulation.
 5. **Protects the environment:** Eco-taxes like carbon taxes promote eco-friendly behavior.
 6. **Market stability:** Keeps businesses stable.
- **Examples:** VAT and income tax in Bangladesh, carbon taxes in India and Europe.

5. Asymmetric Information & Moral Hazard

- **Asymmetric information:** One party knows more than the other, like when a borrower hides financial info from the bank.
- **Moral hazard:** Risky behavior due to unequal information, especially in banking.
 - **Why it's an issue:**
 1. **Borrower's behavior:** Borrowers may take on risky actions, causing losses for banks.
 2. **Economic crisis:** Fraudulent practices lead to financial problems.
 3. **Bank reliability:** Lack of info causes banks to lose money.
 4. **Market instability:** Banks take on too much risk, affecting the whole system.
- **Solution:** Transparency and regulation to ensure trust and fairness.

6. Corrective Taxes: Price Distorting or Correcting?

- **Corrective Taxes** address negative externalities like pollution.
- **Price Correcting:** They adjust prices to reflect the true social cost of production/consumption.
- **Encouraging Efficient Outcomes:** They reduce overproduction/consumption by incorporating external costs.
- **Not Price Distorting:** Unlike regular taxes, corrective taxes help market efficiency by internalizing external costs.

7. Progressive Taxation and Income Distribution

- **Progressive Taxation:** Tax rate increases with income, helping reduce inequality.
- **Reduces Income Inequality:** Wealthier people pay more, funding public services for the less privileged.
- **Wealth Redistribution:** Tax revenues help provide healthcare, education, etc., for the underprivileged.
- **Promotes Economic Justice:** Ensures fairer distribution of wealth.

8. Anti-Trust Regulation

- **Purpose:** Prevents anti-competitive practices like monopolies and price-fixing.
- **Promotes Healthy Competition:** Ensures lower prices, higher quality, and innovation.
- **Encourages Innovation:** Healthy competition drives firms to improve products/services.
- **Protects Consumer Welfare:** Ensures variety and fair pricing.
- **Economic Efficiency:** Prevents market manipulation and encourages resource efficiency.

9. Market Failure in the Financial Sector

- **Reasons for Market Failure:**
 - **Information Asymmetry:** Borrowers often have more information than lenders.
 - **Monopoly Power:** Large banks may reduce competition.
 - **Externalities:** Financial crises can affect broader society.
 - **Imperfect Competition:** A few players can dominate the market.
 - **Lack of Liquidity:** Insufficient liquidity can impair market functioning.
- **Government Interventions:**
 - **Regulation & Supervision:** Ensure transparency and consumer protection.
 - **Monetary & Fiscal Policies:** Stabilize the economy.
 - **Consumer Protection Laws:** Prevent fraud and unethical practices.

10. Negative Externality Diagram

- **Diagram Overview:**
 - **Demand Curve (D):** Represents consumers' willingness to buy at different prices.
 - **Private Supply Curve (S):** Shows costs to the firm without considering external costs.
 - **Social Supply Curve (S'):** Includes the full social cost (e.g., pollution).
 - **Q_{market}:** Equilibrium when only private costs are considered.
 - **Q_{optimal}:** Socially efficient quantity, lower than Q_{market}.
 - **P_{market} & P_{optimal}:** Price shifts higher when external costs are included.
- **Impact:** The diagram shows that corrective taxes or regulation can reduce the negative effects of overproduction and improve efficiency by moving the market to Q_{optimal}.

Module – F

1. Objectives of Macroeconomic Policy

- **Economic Growth:** Increase GDP through development.
- **Inflation Control:** Keep prices stable.
- **Reduce Unemployment:** Create more jobs.
- **Balance of Trade:** Equalize imports and exports.
- **Economic Equality:** Reduce income gap.
- **Social Welfare:** Provide services like health, education.

2. Demand-Pull vs. Cost-Push Inflation

- **Demand-Pull Inflation:**
 - Caused by high demand.
 - AD shifts right → Prices & output rise.
- **Cost-Push Inflation:**
 - Caused by higher production cost.
 - AS shifts left → Prices rise, output falls.

3. Is Inflation Always Bad?

- **Not always.** Low inflation (2–3%) is **good**.
- Helps growth, benefits borrowers.
- But high inflation (10%+) is **harmful**:
 - Reduces purchasing power, savings, and increases inequality.

4. Nominal vs. Real Interest Rate

Type	Explanation
Nominal	Announced rate (e.g., 10%)
Real	Nominal – Inflation (e.g., 10% - 5% = 5%)
Difference	Real shows actual gain; nominal doesn't account for inflation.

5. Seigniorage and Inflation

- **Seigniorage:** Govt earns by printing money.
- More money → higher demand → inflation.
- **If overdone** → Value of money falls → economy suffers.

6. When Is Inflation Beneficial?

- **Low inflation (2-5%)** boosts growth.
- Encourages spending, borrowing, investment.
- Keeps economy active if controlled properly.

7. Bangladesh's Monetary Policy (2024-2025)

- **Interest rate** raised to **10%** in Oct 2024 to fight inflation.
- **Food inflation** still **12%+**, hurting people.
- Govt seeking **\$5 billion** in aid to stabilize economy.
- Policy aims to control inflation, solve **dollar crisis**, and boost **forex reserves**.

8. Selection of Monetary Policy

✓ **Expansionary Monetary Policy** is supported due to:

- **Economic Growth:** Lower interest rates → more investments → growth.
- **Lower Unemployment:** Easier to start businesses → more jobs.
- **New Entrepreneurs:** Easy loans encourage startups.
- **Inflation Control:** Growth stabilizes inflation in the long term.

- **Currency Flow:** More money → active spending and economy.

Example: Bangladesh Bank lowers repo rate to boost economic activity.

9. Causes of Bangladesh's BOP Crisis

Problems:

1. **Trade Deficit** – High imports, low exports.
2. **Falling Remittances** – Less foreign currency inflow.
3. **Inflation** – Higher costs hurt exports.
4. **Global Pressures** – Fuel prices & instability reduce investment.
5. **Reserve Decline** – Pressure on forex reserves.

Solutions:

- Boost exports, support local industries.
- Increase remittances, reform economy.
- Tackle climate impact, build global ties.

Example: RMG exports are not growing fast while oil imports increase.

10. Balance of Payments (BoP)

✓ **Definition:** Record of all international transactions over a period.

Accounts:

1. **Current:** Trade, services, income.
2. **Capital:** Foreign loans, aid.
3. **Financial:** Investments, banking flows.

✓ Why It Balances:

- **Accounting logic:** Every debit has a credit.
- **Deficit compensated** by capital inflow.

Example: If Bangladesh imports more, it may borrow to balance it.

11. National Income

✓ **Definition:** Total value of final goods/services in one year.

Methods:

1. **Production:** Total output.
2. **Income:** Wages + profits + rents + interests.
3. **Expenditure:** $C + I + G + (X - M)$

✦ **Example:** Farmer grows rice (output), sells it (income), buyer consumes (expenditure).

12. Aggregate Demand (AD)

✓ **Definition:** Total demand in an economy at different price levels.

Components: $C + I + G + (X - M)$

Curve: Slopes downward ($\downarrow$ price $\rightarrow$ $\uparrow$ demand).

Reasons:

- **Wealth Effect:** Real wealth $\uparrow$
- **Interest Rate Effect:** Cheap loans $\rightarrow$ $\uparrow$ investment.
- **Net Export Effect:** Lower prices $\rightarrow$ $\uparrow$ exports.

✦ **Example:** Price drop in Bangladesh textiles $\rightarrow$ foreign demand rises.

13. Aggregate Supply (AS)

✓ **Definition:** Total goods/services firms are willing to produce.

SRAS – Upward sloping: Sticky wages, firms produce more when prices rise.

LRAS – Vertical: Long-term output is fixed at full capacity.

Example: In the short run, more garments are produced with price hike, but long run depends on capacity.

14. Interrelation of Macroeconomic Accounts

✓ Four Accounts:

1. **Production** $\rightarrow$ creates **Income**
2. **Income** $\rightarrow$ spent as **Expenditure**
3. **Expenditure** $\rightarrow$ funds flow into **Financial Account**
4. All accounts connect:
 - $\text{Income} = \text{Expenditure}$
 - $\text{Savings} = \text{Investment}$ (Financial flow)

✦ **Example:** A factory produces (production), pays workers (income), who buy goods (expenditure), and invest savings in banks (financial account).

Module – G

1. GNP vs. GDP – Definition & Differences

- **GNP:** Measures income of a country's **citizens**, including income from abroad.
- **GDP:** Measures total production **within the country**, regardless of producer nationality.
- **Key Differences:**
 1. GNP focuses on citizens; GDP focuses on location.
 2. GNP includes income from abroad; GDP excludes it.
 3. GDP includes foreign income inside the country; GNP excludes it.

2. GDP and GNP – Definition, Formula & Key Differences

- **GDP Formula:** $GDP = C + I + G + (X - M)$
- **GNP Formula:** $GNP = GDP + \text{Income from abroad} - \text{Income to foreigners}$
- **Key Point:** GDP is domestic; GNP is citizen-based (regardless of location).

3. Negative Balance of Trade & Bangladesh's External Sector

- **Trade Deficit:** Bangladesh imports more than it exports.
- **Issues:**
 1. **Persistent trade deficit**
 2. **Pressure on foreign reserves**
 3. **RMG-led exports growing but not enough**
 4. **Remittances help offset trade gap**
 5. **Sustainability challenges** but manageable with foreign investment and remittances

4. Changing Trends in Bangladesh's Economic Growth Indicator

1. **GDP Growth:** Steady at 6–7%, led by RMG, services, agriculture.
2. **Industrial Growth:** Strong in RMG, but faces energy & infrastructure issues.
3. **Agriculture:** Improved productivity despite declining GDP share.
4. **Inflation:** Generally controlled; vulnerable to global price shocks.

Short Notes:

1. Economic Systems

An economic system refers to the structure and methods by which a society organizes the production, distribution, and consumption of goods and services. There are four main types of economic systems:

Market Economy: Decisions are made by individuals based on supply and demand.

Command Economy: The government controls the production and distribution of goods.

Mixed Economy: Combines elements of both market and command economies.

Traditional Economy: Based on customs and traditions with limited changes in the production system.

Each system has advantages and disadvantages, depending on the society's goals for economic efficiency, equality, and individual freedoms.

2. Consumer Behaviour

Consumer behavior is the study of how individuals make decisions to spend their resources (time, money, effort) on consumption-related items. It is influenced by psychological, social, and economic factors. Consumers' preferences are influenced by income, tastes, preferences, and the prices of goods and services. It also involves understanding the decision-making process regarding what, when, and how much to purchase. Consumer behavior is essential for businesses to design marketing strategies and optimize product offerings.

3. Marginal Rate of Substitution (MRS)

The Marginal Rate of Substitution (MRS) refers to the rate at which a consumer is willing to exchange one good for another while maintaining the same level of satisfaction or utility. It is represented by the slope of the indifference curve. As a consumer substitutes one good for another, the MRS typically decreases, reflecting diminishing marginal utility. The MRS is crucial in understanding consumer preferences and in deriving demand curves.

4. Budget Line

A budget line represents all possible combinations of two goods that a consumer can purchase given their income and the prices of those goods. It shows the trade-offs a consumer faces when deciding how to allocate their income between different goods. The slope of the budget line is determined by the relative prices of the goods. Any point on the budget line shows an affordable combination, while any point inside the line is a feasible but inefficient allocation.

5. Price Elasticity of Demand

Price elasticity of demand measures the responsiveness of the quantity demanded of a good to a change in its price. It is calculated as the percentage change in quantity demanded divided by the percentage change in price. If demand is elastic, consumers are highly responsive to price changes, while inelastic demand indicates less responsiveness. The elasticity helps businesses set prices and predict consumer behavior in response to price changes.

6. Perfectly Elastic

Perfectly elastic demand occurs when any change in the price of a good results in an infinite change in the quantity demanded. This means that at a specific price, consumers are willing to buy any quantity of the good, but they will buy none if the price changes, no matter how small. Perfectly elastic demand is a theoretical extreme, rarely seen in real life, and is represented by a horizontal demand curve.

7. Perfectly Inelastic

Perfectly inelastic demand is when the quantity demanded of a good does not change regardless of any change in its price. This means that consumers will continue to purchase the same amount of the good even if the price increases or decreases. It is represented by a vertical demand curve. An example could be life-saving medications where consumers must buy the same amount regardless of price changes.

8. Isoquant Curves

Isoquant curves represent all the combinations of two inputs (such as labor and capital) that produce the same level of output. It is similar to an indifference curve but for production instead of consumption. The curve is downward sloping and convex to the origin, reflecting diminishing marginal returns. The shape of the isoquant helps businesses understand the trade-off between different production inputs and the efficiency of their use.

9. Implicit Cost

Implicit costs refer to the opportunity costs of using resources owned by the business owner, where no direct payment is made. These costs are not reflected in accounting profits but represent the foregone income from not using those resources in the next best alternative. For example, the owner's time spent managing the business could have been used in another income-generating activity. Implicit costs are crucial in calculating economic profits.

10. Explicit Cost

Explicit costs are actual out-of-pocket expenses incurred by a business in its production process. These are direct, measurable costs such as wages, rent, and material expenses. Explicit costs are recorded in the financial statements and affect the accounting profits of a business. They are different from implicit costs, which are not recorded in the accounts but still represent real economic costs. Explicit costs are essential in understanding the financial health of a business.

11. Profit Maximization

Profit maximization is the process by which firms determine the price and output level that returns the greatest profit. This occurs when the marginal cost (MC) of producing an additional unit of output equals marginal revenue (MR). Firms aim to produce at the level where $MC = MR$, ensuring they are not overproducing (which would increase costs) or underproducing (which would mean missed opportunities for profit). This concept is central in microeconomics and helps businesses make decisions that maximize their profitability.

12. Productive Inefficiency

Productive inefficiency occurs when an economy or firm produces goods or services at a higher cost than necessary. This often happens when resources are not utilized in the most effective manner, meaning that output could be produced with fewer inputs or at a lower cost. In a productive inefficiency scenario, firms are not operating at the lowest point of their average cost curve, and this reduces overall economic welfare. It typically leads to waste and higher prices for consumers.

13. Allocative Inefficiency

Allocative inefficiency happens when resources are not allocated in the most beneficial way to maximize societal welfare. It occurs when the price of a good or service does not reflect its marginal cost ($P \neq MC$). In perfect competition, allocative efficiency is achieved when firms set the price equal to the marginal cost of production. When allocative inefficiency occurs, it results in the overproduction or underproduction of certain goods, leading to a misallocation of resources.

14. Market Intervention

Market intervention refers to the actions taken by the government or regulatory bodies to influence the functioning of markets. This can involve setting price floors (minimum prices), price ceilings (maximum prices), tariffs, quotas, subsidies, and regulations. The aim of market intervention is often to correct market failures, ensure fair competition, or protect consumers and producers from excessive price fluctuations. However, interventions can sometimes lead to inefficiencies and unintended consequences.

15. Income Distribution

Income distribution refers to the way in which a nation's total income is distributed among its population. It is influenced by factors such as wages, investments, taxation, and government transfers. Inequalities in income distribution can lead to social and economic problems, including poverty, inequality, and reduced access to education and healthcare. Governments often implement policies to redistribute income more evenly through progressive taxation, welfare programs, and social security.

16. Progressive Taxes

Progressive taxes are tax systems where the rate of taxation increases as the income of the taxpayer rises. This means that individuals with higher incomes pay a higher percentage of their income in taxes than those with lower incomes. Progressive taxes aim to reduce income inequality by redistributing wealth from wealthier individuals to those with lower incomes. Common examples include income taxes, where tax rates increase in steps as income rises.

17. Regressive Taxes

Regressive taxes are taxes where the tax rate decreases as the income of the taxpayer increases. In other words, lower-income individuals pay a higher percentage of their income in taxes compared to wealthier individuals. Sales taxes and value-added taxes (VAT) are examples of regressive taxes, as they take a larger proportion of income from lower-income individuals, who spend a larger share of their earnings on goods subject to these taxes.

18. Proportional Taxes

Proportional taxes, also known as flat taxes, are tax systems in which the tax rate remains constant regardless of the taxpayer's income. Everyone pays the same percentage of their income in taxes. For example, a flat tax of 10% would mean that both a person earning \$30,000 and a person earning \$300,000 would pay 10% of their income in taxes. This type of tax is simpler to administer but may not reduce income inequality as effectively as progressive taxes.

19. Transfer Payments

Transfer payments are payments made by the government to individuals without any goods or services being provided in return. These payments are typically aimed at reducing inequality or providing financial support to vulnerable groups, such as the elderly, unemployed, or low-income individuals. Examples include unemployment benefits, pensions, and welfare payments. Transfer payments help in income redistribution and support individuals during times of economic hardship.

20. Aggregate Demand

Aggregate demand (AD) represents the total quantity of goods and services demanded in an economy at a given overall price level and within a specific period. It is the sum of consumption, investment, government spending, and net exports (exports minus imports). The aggregate demand curve typically slopes downward, meaning that as the price level falls, the quantity of goods and services demanded increases. Changes in any of the components of AD can lead to shifts in the curve, affecting overall economic output.

21. Aggregate Supply

Aggregate supply (AS) is the total quantity of goods and services that producers in an economy are willing and able to produce at different price levels in a given time period. The aggregate supply curve can be vertical, horizontal, or upward sloping depending on the time horizon (short-run or long-run). In the short-run, AS can be influenced by factors like input prices and wage rates, while in the long-run, it is determined by factors such as technological advancements and capital stock.

22. Business Cycle

The business cycle refers to the fluctuations in economic activity that an economy experiences over time, typically consisting of four phases: expansion, peak, contraction (recession), and trough. During expansion, economic output increases, leading to higher employment and growth. The peak represents the point of maximum economic output, after which a contraction occurs, resulting in a decline in economic activity. The trough marks the lowest point, after which the cycle may begin to expand again.

23. Supply-side Policies

Supply-side policies are aimed at increasing the overall supply of goods and services in an economy. These policies focus on improving the efficiency and productivity of production by reducing barriers to production, such as high taxes, excessive regulation, and labor market restrictions. Examples include tax cuts for businesses, deregulation, and policies to improve education and workforce skills. The goal of supply-side policies is to increase long-term economic growth by encouraging investment and production.

24. Fiscal Accounts

Fiscal accounts refer to the record of a government's revenue and expenditure, which reflects its fiscal policy. These accounts include tax revenues, government spending on goods and services, and transfers. Fiscal accounts are essential for assessing the government's financial health and sustainability. A fiscal surplus occurs when government revenues exceed expenditures, while a fiscal deficit occurs when expenditures surpass revenues. Fiscal policies can influence national savings, investment, and overall economic stability.

25. Monetary Accounts

Monetary accounts track the supply and demand for money within an economy. This includes the central bank's management of monetary policy, interest rates, and the money supply. The monetary accounts aim to control inflation, stabilize the currency, and promote economic growth. Key tools used include setting reserve requirements for banks, engaging in open market operations, and adjusting the benchmark interest rates. Monetary policy works in tandem with fiscal policy to manage an economy's performance.

26. Per Capita GDP

Per capita GDP is a measure of the average economic output per person in a country. It is calculated by dividing a country's total Gross Domestic Product (GDP) by its population. This indicator helps to compare the standard of living between countries or regions. While it provides a useful snapshot of average income levels, it does not account for income inequality or differences in cost of living across regions.

27. GNI (Gross National Income)

Gross National Income (GNI) is the total income earned by a country's residents both domestically and internationally, including income earned by citizens working abroad or through investments. It is the sum of GDP and net income from abroad (such as income from foreign investments). GNI is a broader measure of a country's economic health than GDP, as it includes international income flows. It is often used to assess the economic well-being of a nation and to compare economies globally.